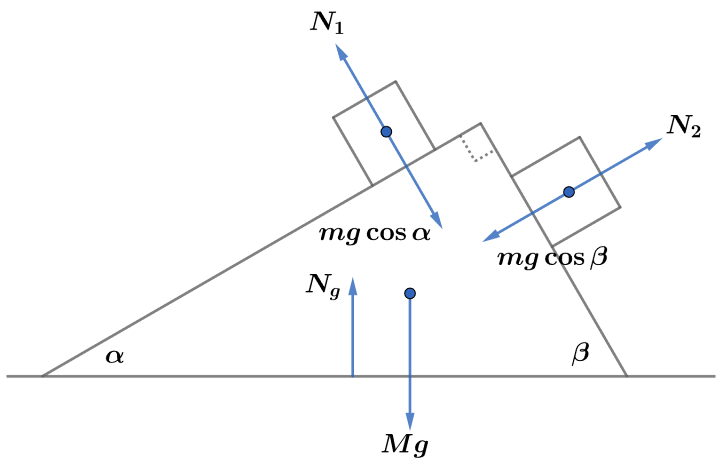
2013 F=ma Exam: Problem 11

Kevin S. Huang



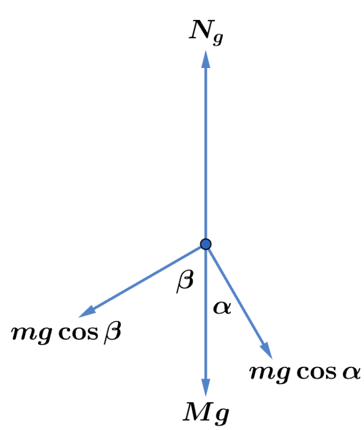
Since the triangular block remains at rest, the net force on it is zero. In addition to its own weight, the block feels the normal forces from the two cubes and the ground. As the cubes are effectively sliding down an incline, the normal force on the left cube has magnitude

$$N_1 = mg\cos\alpha$$

while the normal force on the right cube has magnitude

$$N_2 = mg\cos\beta$$

By Newton's 3rd law, the normal forces from the cubes on the block have the same magnitude in the opposite direction. Thus, the free-body diagram for M is as follows:



Balancing forces in the vertical direction,

$$N_g = Mg + mg\cos^2\alpha + mg\cos^2\beta$$

Since $\beta = \frac{\pi}{2} - \alpha$,

$$\cos\beta = \cos\left(\frac{\pi}{2} - \alpha\right) = \sin\alpha$$

Since $\sin^2\theta + \cos^2\theta = 1$,

$$N_g = Mg + mg\cos^2\alpha + mg\sin^2\alpha = Mg + mg$$

so the answer is $\boxed{\text{C}}$.